OPOCA: One Point One Class Annotation for LiDAR Point Cloud Semantic Segmentation

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***Abstract*— This article tackles the problem of requiring a large amount of data annotation in the LiDAR point cloud semantic segmentation (PCSS) task by proposing OPOCA, a weakly supervised network that only annotates one point per class in a single LiDAR scan. To compensate for the supervisory losses due to extremely few annotated labels, a large number of pseudo labels is first generated using a pseudo label spreading module (PLSM), whereas the potential ambiguity and inaccuracy are further addressed by a carefully-designed spread distance loss (SDL) and a range image auxiliary module (RIAM). Moreover, we propose an iterative self-training module (STM) to increase the high-quality pseudo labels for the next round of training. Extensive experiments on various benchmark datasets (SemanticKITTI, Waymo Open Dataset (WOD), and SemanticPOSS) demonstrate the rationality of each module and the superior performance of the proposed network over the current baseline with below 0.1‰ labels.**

***Index Terms*— Light detection and ranging (LiDAR), one point one class annotation (OPOCA), point cloud, semantic segmentation, weakly supervised.**

I. Introduction

LIGHT detection and ranging (LiDAR) point cloud semantic segmentation (PCSS) is a fundamental task for fine-grained scene understanding with applications ranging from remote sensing [1], robotics [2], and self-driving [3]. Compared with images from cameras, LiDAR point clouds collected from LiDAR sensors can capture accurate geometric and depth information [4], [5], [6].

Over the past years, full-supervised learning methods [7], [8] have achieved excellent performance due to the emergency of large-scale annotation datasets, such as SemanticKITTI [9] and Waymo Open Dataset (WOD) [10]. However, labeling LiDAR point clouds on these datasets is time-consuming and labor-intensive work. One possible solution is to fully exploit the label information, including the geometry patterns in scenes [11], local semantic similarity features [12], and proxy tasks [13]. Therefore, some weakly supervised PCSS methods [14], [15] are proposed to enhance supervision signals by generating pseudo-labels. Nevertheless, the annotation of LiDAR point clouds in them presents several challenges.

1) The massive labeled data is still needed due to a large number of collected points, although they typically require 10% to 0.1‰ labels. For example, SemanticKITTI [9] has more than 2×10^9 points and WOD [10] has more than 3.7×10^9 points.
2) The random sampling strategy is used to pick points to annotate, causing some categories to be missing.
3) Utilizing random sampling for annotation on highly sparse LiDAR point clouds proves unfeasible due to a sharp decline in accuracy when annotation counts dip below a particular threshold.

To address the above challenges, we propose a novel weakly supervised (PCSS) method based on a one point one class annotation strategy, named OPOCA, aiming to reduce the demands for annotation labor and time and avoid the side effects due to the random sampling annotation strategy (see Fig. 1). The key to necessitating sparse labeling is to involve only a solitary point per class within each LiDAR scan. Compared with the annotations in fully supervised and previous weakly supervised PCSS methods, our OPOCA only needs below 0.1‰ point labels and uniformly annotates all categories. Specifically, we first propose a pseudo label spreading module (PLSM) to provide more pseudo labels, aiming to compensate for the supervisory losses due to extremely little annotations. PLSM filters out the ground points, clusters the objects, and spreads semantic labels in a reasonable way. One problem with the PLSM is that naively spreading pseudo labels would have a misleading effect due to the potential ambiguity and inaccuracy in the pseudo labels. Therefore, we propose to introduce a carefully designed spread distance loss (SDL) and a range image auxiliary module (RIAM). The proposed SDL can explicitly spread labels correctly by assigning different weights based on the distance between the neighbors. The designed RIAM further reduces the misleading effect by incorporating the range of images transformed from LiDAR point clouds. Moreover, an iterative self-training module (STM) is introduced to increase generated pseudo labels with high confidence in the training stage.

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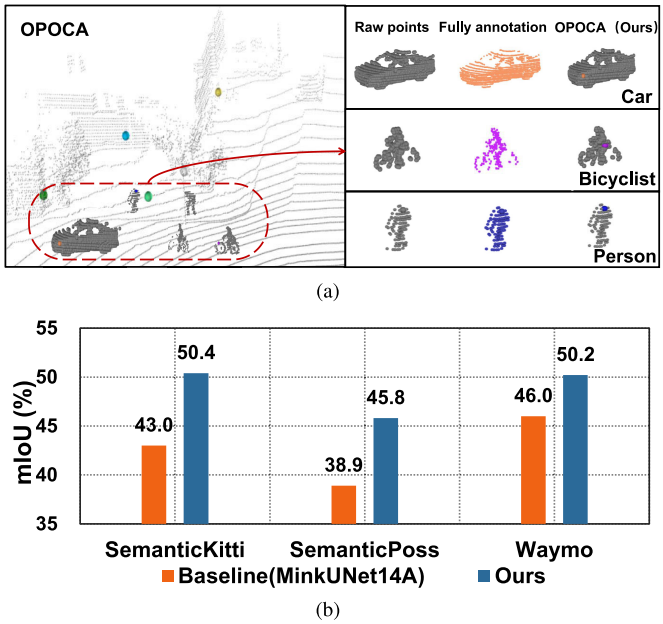


Fig. 1. (a) Examples of our OPOCA strategy. (Left) A local region in one LiDAR scan annotated by OPOCA, only one point per class is annotated. (Right) The number of annotations drastically reduce comparing to full annotation in the class car, bicyclist, and person. (b) Performance comparison between baseline and our OPOCA on various benchmarks. The proposed OPOCA delivers better performance with below 0.1‰ labels.

To summarize, the main contributions of this work are as follows.

1) We propose a weakly supervised LiDAR PCSS method OPOCA, that employs an OPOCA strategy. OPOCA significantly reduces the difficulty of annotation and the requirement for the number of labels.
2) We propose a PLSM to generate high-quality pseudo labels, compensating for the supervisory losses due to very few annotations. Based on these pseudo labels, we further propose a carefully-designed SDL and a RIAM to reduce the potential ambiguity and inaccuracy in the generated pseudo labels.
3) We propose an iterative STM to gradually increase the number of pseudo-labels with high confidence for the next round of training.
4) We conduct extensive experiments on various benchmarks, including SemanticKITTI [9], WOD [10], and SemanticPoss [16], to demonstrate the superior performance of our OPOCA over the current baselines.

II. Related Work

A. Point Cloud Feature Learning

Point cloud learning methods can be categorized into point-based, voxel-based, and image-based according to the different input representations [17], [18], [19], [20]. To handle structurally unordered point cloud data, pioneering point-based methods [21], [22] introduce a global-max operator to aggregate learned features. Furthermore, several variants of point-convolutional operations [8], [23] have been proposed for feature extraction. However, the point-based methods need a significant amount of processing time to find neighborhoods. On the other hand, the voxel-based methods [24], [25] first translate point clouds into regular voxelized grids and then use traditional 3-D convolution techniques. However, 3-D convolution has growing computational cubic complexity, which imposes a resolution constraint. To solve it, sparse convolution techniques [26], [27], [28], which operate only on nonempty indices, can significantly reduce computational complexity and improve learning efficiency. Image-based methods [29], [30] project point clouds onto range images using index mapping and then leveraging well-developed 2-D feature extraction strategies. However, postprocessing is required since range image projection inevitably leads to a loss of spatial information.

Recently, combining multiple input representations to overcome the limitations of a single one is also proposed [31], [32]. RPVNet [33] combines features from the point, voxel, and range image branches, which are then reprojected onto the corresponding views, enhancing the fusion of multiple views. Some works [34], [35] combine point-based and voxel-based representations for reducing information loss during voxelization. Inspired by them, in this work, we incorporate the range image features with the voxel-based network to enhance the overall effectiveness of the labeling strategy.

B. Weakly Supervised PCSS

Given incomplete and sparse annotations, weakly supervised methods draw increasing attention to PCSS tasks. Various sparse labeling strategies have been proposed, including random labeling [12], [36], scribble labels [14], bounding-box annotations [37], object-level annotations [38], scene or subcloud level labels [15], [39], active annotations [40], [41], and interval frame annotations [42].

The most common approach to weak supervision involves random labeling based on a specific proportion [12], [36]. However, the uniform distribution of these randomly annotated points often diverges significantly from manual annotations. ScribbleKITTI [14] employs scribble labels for training, which are more efficient than point-by-point annotation, yet the annotated points still represent a significant portion (8.06%) of the total data. Box2Mask [37] uses bounding box labels as a more cost-effective method compared to point-wise annotation, but this requires careful consideration of both the position and size of the box. OTOC [38] mandates annotating a point for each object, which can be costly in outdoor scenes where a single class might include multiple objects, thereby multiplying the annotation effort compared to class-based annotation strategies. MPRM [15] leverages scene/subcloud-level labels and utilizes the class activation map (CAM) to identify categories. However, CAM encounters challenges in distinguishing nondiscriminative regions of objects. Shao et al. [40] and Hu et al. [41] employed various sample selection methods for active annotation. Despite their efficiency, they necessitate multiple rounds of manual annotation, adding complexity to the annotation process. Recent methods LESS [11] and OCOC [43] have tried to segment the point cloud into regions and then annotate one point per class in each region.

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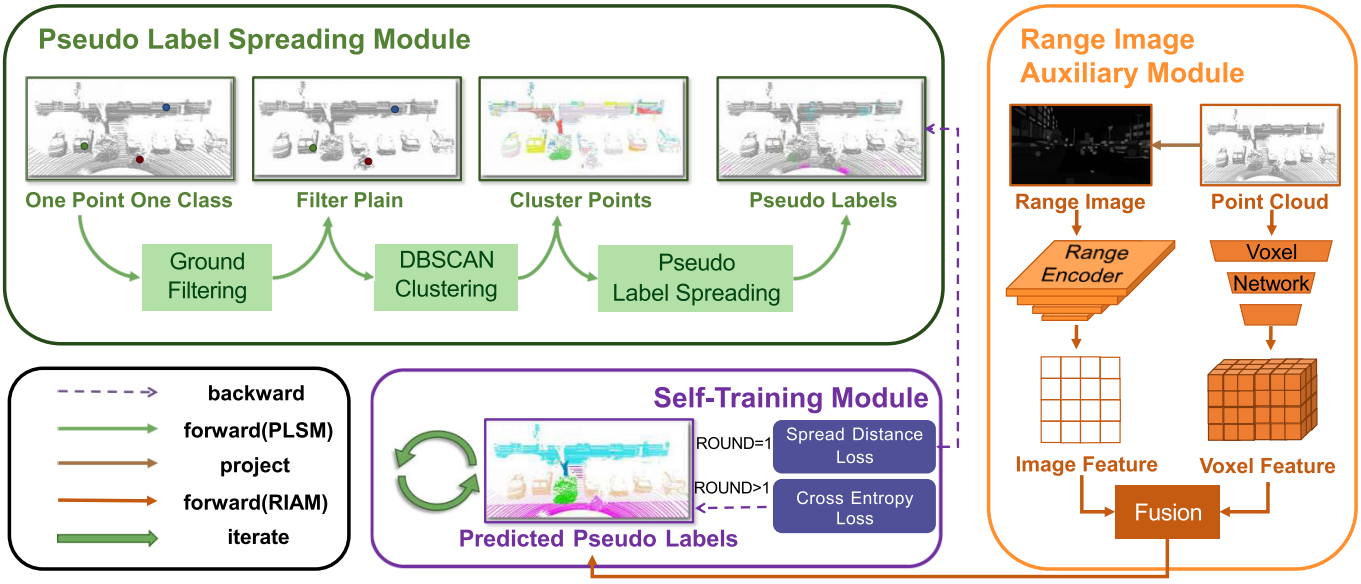


Fig. 2. Illustration of our weakly supervised LiDAR PCSS method (OPOCA). The PLSM first generates pseudo labels based on the extremely few annotation labels. The potential ambiguity and inaccuracy in PLSM are then addressed by an SDL and a RIAM. Finally, the STM combines a predicted and confidence filter to generate pseudo-labels that are used for the next round of training.

Both are different from our method; our method directly labels the raw point cloud. This eliminates the need for a preliminary segmentation step. In addition, our OPOCA reduces the cost by conducting annotations at the frame level.

III. Methodology

Fig. 2 shows the network architecture of our OPOCA. The PLSM first generates pseudo labels based on the extremely few annotated labels. The potential ambiguity and inaccuracy in PLSM are then addressed by an SDL and a RIAM. Finally, the STM combines a predicted and confidence filter to generate pseudo-labels that are used for the next round of training. For simplicity in the subsequent discussion, we use p_i to refer to the ith point in the point cloud, and l_i to denote the label of point p_i. P_L denotes the set of points with manually assigned labels, while P_{psu} represents the set of points with pseudo-labels.

A. Pseudo Label Spreading Module

To address the insufficient supervisory information due to extremely sparse annotations in OPOCA, we propose the PLSM (see Fig. 3), which aims to generate a number of pseudo labels. PLSM first separates the ground points from other points since the ground is not always perfectly level throughout a LiDAR scan. We split a LiDAR point cloud into subregions based on the $x-y$ plane and apply the RANSAC algorithm [44] in each subregion that contains a labeled point to separate the ground points. After filtering out the ground points, we adopt the hypothesis that points in a local region have similar semantics. Next, we use the DBSCAN algorithm to cluster the objects based on the hypothesis that the points in the same set should have the same semantics.

However, the distance between local points is relatively large due to the sparsity of LiDAR scans, leading to less

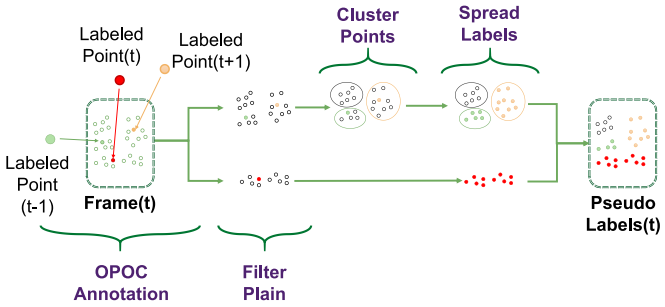


Fig. 3. Architecture of the PLSM.

accurate clustering results. To address this problem, we further introduce the normal vector constraints to ensure the accuracy of label propagation. The normal vectors of all points within the set and the normal vector distance between them and the labeled point p_i are computed. We only consider the points within the set whose normal vector distance to p_i is close enough to be of the same semantic category. Consequently, we assign the semantic label l_i to the points within the set that contains point $p_i \in P_L$.

We examine the points in P_{psu}, where semantic labels are propagated from P_L, leading to a significant increase in the number of labeled points. However, P_{psu} and P_L are concentrated in only a few local areas, that lack spatial diversity. LiDAR point cloud sequences are often captured for auto-driving scenes, The selection of points in the OPOCA is also random. Therefore, the labeled points, P_L, in different scans are usually not in the same local area in the current scan. Thus, so that the OPOC labels in adjacent scans may occur in different regions in the current scan, we use calibration parameters to register adjacent scans to the current scan. This allows us to obtain the labeled points in P_L from adjacent scans and then spread the adjacent labels in the current frame, thereby increasing spatial diversity.

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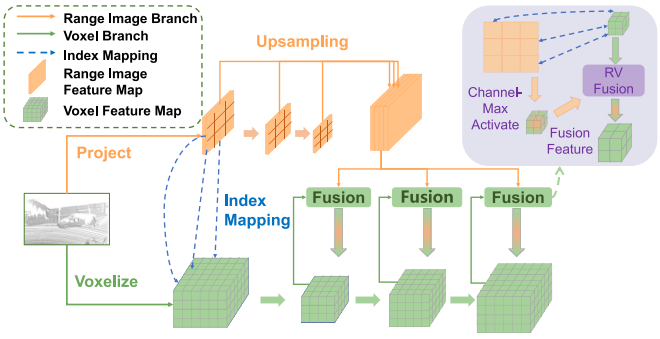


Fig. 4. Architecture of the RIAM, and the voxel network is our backbone.

B. Spread Distance Loss

The proposed SDL aims to reduce the impact of mislabeling in the erroneous pseudo-labels obtained through PLSM. The loss consists of two components

$$\text{Loss} = \text{Loss}_{\text{gt}} + \text{Loss}_{\text{psu}}. \tag{1}$$

Loss_{gt} is the loss for the points with ground truth labels; N_{gt} represents the number of those points; y and $\hat{y}$ represent the ground truth and prediction, respectively. These points are trained using the following commonly used cross-entropy loss:

$$\text{Loss}_{\text{gt}} = -\frac{1}{N_{\text{gt}}} \sum_{i=1}^{N_{\text{gt}}} \sum_{c=1}^{K} y_{ic} \log \hat{y}_{ic}. \tag{2}$$

On the other hand, Loss_{psu} is the loss for the points that have pseudo labels from PLSM, and the number of these points is represented by N_{psu}. Based on this, the spread distance is proposed to employ the pseudo labels selectively. Specifically, for each point p_i in P_L that spreads its labels to a set of points P_{psu_i}, we calculate the Euclidean distance between p_i and all points in P_{psu_i}, denoted as D_i. Each point p_j in P_{psu_i} is assigned a weight, w_j, based on its distance d_j from p_i. The weight is calculated as

$$w_j = 1 - \frac{d_j}{\max\{D_i\}}, \quad d_j \in D_i. \tag{3}$$

The loss for points with pseudo labels is then combined with their respective weights as

$$\text{Loss}_{\text{psu}} = -\frac{1}{N_{\text{psu}}} \sum_{j=1}^{N_{\text{psu}}} w_j \sum_{c=1}^{K} y_{jc} \log \hat{y}_{jc}. \tag{4}$$

Incorporating the SDL into the overall loss function reduces the impact of mislabeling in the erroneous pseudo labels.

C. Range Image Auxiliary Module

To leverage information from various perspectives, we introduce the RAIM (see Fig. 4), which takes range images as input to enhance feature extraction capability. Simultaneously, we extract the most salient features from OPOC labels, thereby improving the discriminative ability of semantic features. Specifically, our pipeline does not impose any specific restrictions on the choice of the range image encoder. For efficiency, we employ bilinear interpolation on the feature layers of different resolutions in the range image encoder. This allows us to obtain feature maps of multiple input resolutions and then concatenate these feature maps to create a single feature map that integrates all channels. To establish the mapping between the range image feature map and the feature maps of different layers in a voxel-based backbone, the index mapping is preprocessed based on the operator to generate the range image from the LiDAR point cloud. Then, the range image features are projected to the corresponding voxels. For multiple range image features projected to the same voxel, the channel-max operation is applied to activate the feature with the highest value. Next, we input the features from the voxel network and the index mapping-based range image features into the feature fusion module, which performs by using a multilayer perceptron (MLP). As a result, we obtain a feature map with the same dimensions as the voxel feature in the voxel network structure. Finally, the features of each layer in the encoder are replaced by the fused features to leverage additional information.

Algorithm 1 Self Training Module (STM)

Input LiDAR Point Cloud **P** and One Point One Class Annotation **L**
procedure
  # The Model Represents RIAM-embedded Backbone
  # Train The First Model
  $\mathbf{L_{psu}} \leftarrow$ Pseudo Label Spreading Module(**L**)
  $\mathbf{Model_0} \leftarrow$ Train with Spread Distance Loss(**P**, $\mathbf{L_{psu}}$)
  # Self Training Stage
  for i $\leftarrow$ 1 to N by 1 **do**
    $\mathbf{L_{i0}} \leftarrow \mathbf{Model_{i-1}}$(**P**) predict using **TTA**
    $\mathbf{L_i} \leftarrow$ Confidence Filter ($\mathbf{L_{i0}}$)
    $\mathbf{Model_i} \leftarrow$ Train with Cross-Entropy Loss(**P**, $\mathbf{L_i}$)
  end for
  $\mathbf{Model} \leftarrow \mathbf{Model_N}$
end procedure

D. Self-Training Module

To further increase the number of high-quality labels. We propose STM, which is able to use the current predicted labels as the new generation of pseudo labels for the next round of training, thus providing more high-quality labels.

To improve the accuracy of pseudo label generation, the test-time augmentation (TTA) [45] strategy is applied. This strategy involves applying random transformations to the LiDAR point cloud, such as rotating it by a random angle, inverting the plane with a certain probability, and adding random noise multiple times. Then, the predicted semantic segmentation labels are acquired by aggregating the predictions of these multiple transforms.

Finally, the predicted labels with high confidence are considered as new pseudo labels, which serve as the supervision signal for the next training iteration together with the ground truth labels. By applying such a scheme iteratively, the number of high-quality pseudo-labels gradually increases. The detailed procedure is provided in Algorithm 1.

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IV. Experiments

In this section, we first describe the experimental settings. Next, we present experiments on three benchmarks SemanticKITTI, WOD, and SemanticPOSS to demonstrate the effectiveness of our OPOCA.

A. Experimental Setup

1) Dataset Setting: For SemanticKITTI [9] dataset, the valid LiDAR point cloud XYZ-space was set from [−50, −50, −4 m] to [50, 50, 2 m] for both training and validation. The LiDAR point cloud was voxelized with a voxel size of 0.05 m. As for the WOD [10], we set the valid LiDAR point cloud XYZ-space from [−75, −75, −2 m] to [75, 75, 4 m], following the setup commonly used in object detection. The voxel size was set to [0.075, 0.075, 0.05 m]. Some frames in WOD [10] have a large number of points; thus, to manage memory constraints, we randomly downsampled the LiDAR point cloud to 120k points during training. Regarding SemanticPOSS [16], the valid point range was set from [−75, −75, −4 m] to [75, 75, 4 m], and the voxel size was set to [0.075, 0.075, 0.05 m]. For the RIAM, we projected the LiDAR point cloud from all three datasets onto a range image with the size of [2048, 64], thereby avoiding the need to migrate models across different datasets.

2) Training Details: For PLSM, several hyperparameters must be set manually. For RANSAC, we set the threshold for the maximum distance a point can be from the plane model to 0.129. The number of initial points to be considered inliers in each iteration is 3, and the number of iterations is set to 40. When using DBSCAN for clustering, the density parameter used to find neighboring points is set to 0.1, and the minimum number of points required to form a cluster is set to 5. However, please note that these parameters are roughly set based on experience and have not been carefully tuned, so they may not be the optimal choices. Also, due to the large proportion of ground points, when further constraining the clustering results based on the cosine similarity of the normal vectors, the similarity threshold for the ground category is set to 0.99, while the threshold for other categories is set to 0.95.

In our experiments, we trained the models using the SGD optimizer with a learning rate of 0.24 for 50 epochs. The cosine annealing learning rate strategy was adopted for learning rate decay, and all experiments were trained from scratch. To ensure a fair comparison and accommodate GPU memory limitations, we used a batch size of 8 for the baseline model and a batch size of 4 for the combination of RIAM. During training, we applied widely used data augmentation strategies for segmentation. These included global scaling with a random scaling factor sampled from the range [0.95, 1.05], global rotation around the Z-axis with a random angle, and random flips along the x, y, or $x + y$ planes. The TTA data augmentation method in the STM predictive pseudo labels also used the same three augmentation strategies and the parameter for TTA is set to 12. For SemanticKITTI, our PLSM module required approximately 36 h for processing. Each round of iterative training took about 50 h, and self-training inference prediction with TTA = 12 took about 4 h. To save time, for self-training, we trained only one round.

3) Other Setting and Dependency: We conducted all our experiments on Ubuntu 16.04 using 1 NVIDIA GeForce RTX 3090 GPU and the CPU has a clock speed of 2.5 GHz and 48 cores. To implement our models, we used the PyTorch deep learning framework [46] and the sparse convolution framework SpConv [27]. For PLSM processing, we used the 3-D data processing library Open3D [47]. In addition, we efficiently implemented index mapping in the RIAM using TorchSparse [28].

It should be noted that our method does not impose any restrictions on the specific backbone network, and we have chosen not to use more advanced and resource-intensive models due to considerations of training efficiency and computational resources. Instead, we have utilized the efficient and well-performing minkUnet14A as the backbone for all experiments, with our main focus being on the improvements compared to the baseline.

4) Evaluation Metric: Given the 3-D PCSS task, we adapted the most widely used mean intersection over union (mIoU) as the evaluation metric. This metric is calculated using the formula

$$\mathrm{IoU}_c = \frac{\mathrm{TP}_c}{\mathrm{TP}_c + \mathrm{FP}_c + \mathrm{FN}_c} \tag{5}$$

where TP_c, FP_c, and FN_c represent the number of true positive, false positive, and false negative predictions of class c, respectively. The mIoU is then obtained by taking the average over all the classes

$$\mathrm{mIoU} = \frac{1}{K}\sum_{c=1}^{K} \mathrm{IoU}_c. \tag{6}$$

In this formula, K represents the number of classes considered.

B. Results on SemanticKITTI

In this experiment, we use the MinkUNet14A [26] backbone as the baseline for comparing our OPOCA with existing fully supervised methods and random percentage labels. The number of OPOCA in SemanticKITTI accounts for only 0.092‰ of the total number of points, which is slightly less than the number of randomly sampled 0.1‰. The effectiveness of our method is measured by the improvement compared to the baseline. As shown in Table I, compared to full supervision, utilizing only 1% or 0.1% of the labels results in only a marginal decrease in performance. However, under the labeling strategy of 0.01% and OPOC, there is a notable decline in accuracy. This highlights the challenges posed by OPOC labeling and indicates that numerous existing methods are not well-suited to address annotations of such sparsity. And in Table I, our OPOCA, which improves the baseline by **7.4** mIoU, demonstrates the effectiveness of our approach. Out of the 19 classes, 16 classes exhibit improvement compared to the baseline, illustrating the broad validity of our OPOCA. In addition, our method achieves comparable or even better accuracy than some methods that employ 0.1‰ random annotation. Fig. 5 shows the qualitative comparisons of ground truth, fully supervised methods, the baseline, and our OPOCA.

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TABLE I

COMPARISON RESULTS ON OFFICIAL SEMANTICKITTI SINGLE SCAN VALID DATASET. OPOC MEANS ANNOTATING ONE POINT FOR ONE CLASS IN SINGLE LIDAR SCANS. † REPRESENTS THE REPRODUCED RESULT

Setting	Model	mIoU	car	bicycle	motorcycle	truck	other-vehicle	person	bicyclist	motorcyclist	road	parking	sidewalk	other-ground	building	fence	vegetation	trunk	terrain	pole	traffic-sign
Full.Sup.	MinkowskiNet [26]	61.1	95.7	20.4	63.9	70.3	45.5	65	78.5	0	93.5	49.6	81	0.2	**91.1**	**63.8**	87.2	68.5	72.3	64.4	49.1
	SPVCNN [34]	63.8	97.1	35.2	64.6	72.7	64.3	69.7	82.5	0.2	93.5	**50.8**	81	0.3	**91.1**	63.5	**89.2**	66.1	**77.2**	64.1	49.4
	Cylinder3D [24]	**65.9**	**97.2**	**54.8**	**78.9**	**79.6**	**67.1**	**75.4**	**86.2**	1.0	**94.7**	45.6	**81.7**	0.6	89.3	53.0	87.0	**70.8**	71.4	**65.7**	**52.8**
	HybridCR [36]	54.0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	MinkUNet14A† [26]	60.6	89.8	34.0	72.9	65.8	56.3	73.3	79.1	**1.8**	93.0	44.8	79.8	**3.2**	85.9	46.8	85.6	60.6	74.3	58.3	46.4
1%	HybridCR [36]	52.3	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	MinkUNet14A† [26]	**59.2**	**88.8**	**30.6**	**60.4**	**81.8**	**51.4**	**59.2**	**84.1**	0.0	**93.0**	**44.1**	**79.8**	**1.5**	**85.7**	**45.6**	**85.6**	**62.6**	**74.6**	**55.5**	**41.4**
0.1%	MinkUNet [26]	55.5	95.0	0.9	47.4	64.5	46.4	45.3	70.3	**0.3**	92.5	43.0	79.0	0.9	88.7	53.4	87.3	63.4	73.7	59.9	43.2
	SQN(MinkUNet) [12]	55.8	94.9	5.7	43.2	66.8	43.6	49.5	66.3	0.0	91.5	41.1	78.0	0.9	88.5	55.9	**88.2**	64.0	**75.5**	61.1	45.2
	LESS(Cylinder3D) [11]	**66.0**	**97**	**50**	**73**	**94**	**67**	**76**	**92**	0	**93**	40	79	3	**91**	**60**	87	**68**	71	**62**	**51**
	MinkUNet14A† [26]	56.6	88.3	27.3	55.9	66.9	43.1	52.0	78.5	0.2	92.3	**43.7**	**79.3**	**4.0**	85.3	45.3	85.4	64.2	74.3	53.2	37.2
0.01%	MinkUNet [26]	43.2	92.4	0.0	24.8	25.8	20.1	9.6	15.9	0.0	89.3	32.1	74.8	0.0	87.6	**55.1**	87.1	56.4	73.2	48.2	28.3
	SQN(MinkUNet) [12]	50.0	93.5	0.2	35.6	47.2	31.6	33.8	59.1	0.0	89.7	31.9	75.6	0.2	87.6	52.9	**88.2**	58.0	**76.0**	52.1	36.4
	LESS(Cylinder3D) [11]	**61.0**	**96**	**33**	**61**	**73**	**59**	**68**	**87**	0	**92**	**38**	**76**	**5**	**89**	52	87	**67**	71	**59**	**46**
	MinkUNet14A† [26]	42.7	86.3	1.1	23.6	38.4	25.6	32.1	0.0	0.0	88.0	29.6	73.9	1.1	82.2	43.6	84.9	57.7	74.5	47.9	22.0
OPOC	SQN [12]	34.4	84.9	3.1	11.4	17.2	17.1	16.7	34.9	0.0	74.9	15.8	57.3	0.7	77.9	23.3	70.8	35.2	70.2	30.6	11.5
	Baseline(MinkUNet14A†) [26]	43.0	**86.3**	12.6	26.9	38.1	26.5	24.0	52.4	**0.1**	88.4	24.6	73.5	1.4	82.0	33.6	78.6	42.9	72.6	**38.1**	13.8
	Ours(MinkUNet14A†)	**50.4**	85.5	**23.3**	**53.0**	**66.2**	**35.4**	**39.0**	**62.3**	0.0	**91.3**	**36.4**	**78.0**	**4.8**	**84.8**	**37.3**	**82.7**	**49.4**	**74.3**	28.4	**25.8**

TABLE II

COMPARISON RESULTS ON THE WOD. † REPRESENTS THE REPRODUCED RESULTS. SF, SC, AND MU DENOTE SPHEREFORMER [7], SPARSECONV [48], AND MINKUNET14A [26], RESPECTIVELY. OPOC MEANS ANNOTATING ONE POINT FOR ONE CLASS IN SINGLE LIDAR SCANS

Setting	Model	mIoU	car	truck	bus	other vehicle	motorcyclist	bicyclist	pedestrian	sign	traffic light	pole	construction cone	bicycle	motorcycle	building	vegetation	tree trunk	curb	road	lane marker	other ground	walkable	sidewalk
Full.Sup.	SF	**69.9**	**94.5**	**61.6**	**87.7**	**40.2**	0.9	**69.7**	**90.2**	**73.9**	**41.8**	**77.2**	**65.4**	**71.9**	**83.7**	**95.9**	**91.7**	**68.4**	**69.8**	**93.3**	**53.9**	**47.9**	**80.8**	**77.2**
	SC	66.6	94.4	59.8	85.1	37.8	**2.2**	69.1	89.3	73.4	40.4	74.8	57.3	66.6	75.2	95.5	91.3	67.0	68.1	92.3	41.7	30.1	79.0	75.6
	MU†	59.8	92.3	55.6	75.8	29.7	0.4	61.6	78.1	64.2	32.8	65.1	57.6	52.3	56.1	91.4	88.5	61.3	62.6	88.4	36.6	22.5	72.6	69.3
OPOC	Baseline(MU†)	38.9	89.1	34.2	46.7	11.1	**0.5**	33.7	51.9	45.8	14.1	**53.0**	20.2	14.1	10.8	86.1	81.1	**45.6**	**41.0**	64.1	7.5	3.6	51.5	49.9
	Ours(MU†)	**45.8**	**90.2**	**52.7**	**65.7**	**21.3**	0.0	**41.5**	**63.0**	**49.5**	**25.4**	50.3	**31.5**	**17.8**	**30.5**	**89.3**	**82.3**	44.2	37.5	**73.7**	**15.0**	**11.4**	**58.9**	**55.3**

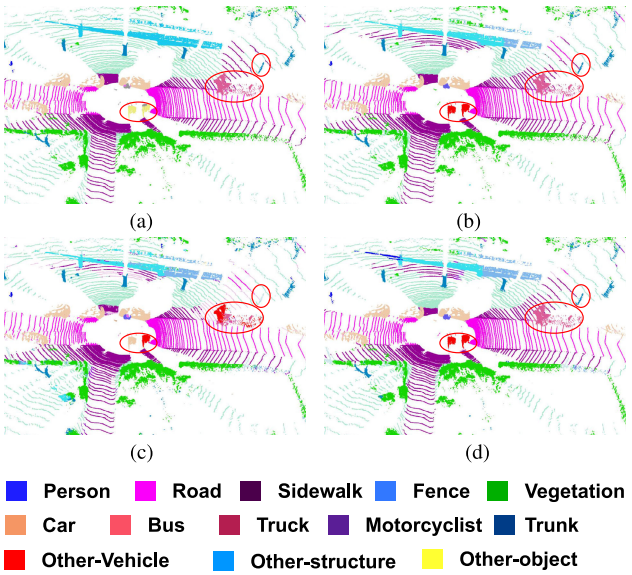


Fig. 5. Qualitative results achieved by (a) ground truth, (b) full-supervised, (c) baseline, (d) our OPOCA on the SemanticKITTI dataset.

As seen in Table I, LESS [11] surpasses us since it adopts the backbone Cylinder3D [24], achieving the better segmentation performance (65.9 mIoU) than our MinkUNet14A (60.6 mIoU) in the full supervision on the SemanticKITTI dataset. However, Cylinder3D is unaffordable for us due to resource limits (a single full experiment will be demanded over 20 days). In addition, annotations in LESS are region-based, making the count of annotations several times our frame-based method. In the OPOC setting, SQN's performance experiences a significant decline compared to 0.01% random annotation, possibly due to its reliance on a uniform labeled point distribution.

During the inference stage, there is no need to consider PLSM and STM. In the efficiency experiments on the SemanticKITTI validation set, our method exhibits an approximate inference time of 89.469 ms/frame.

C. Results on WOD

In this experiment, we compared the performance of our OPOCA with existing fully supervised methods and the MinkUNet14A backbone [26] as the baseline. In WOD, the number of OPOCA accounts for approximately 0.090‰ of the total points. This is marginally lower than the count of a random sample comprising 0.1‰ of the points. The results (see Table II) indicate that our OPOCA outperforms the baseline by **6.9** mIoU. Furthermore, the accuracy of all 18 out of the 22 classes has improved to varying degrees. These findings demonstrate that our method effectively enhances the accuracy of the majority of categories and achieves significant overall improvement. The qualitative results (see Fig. 6) further support the effectiveness of our OPOCA. The comparison between the ground truth, fully supervised methods, the baseline, and our method demonstrates the improvements achieved by our method. Overall, the experimental and qualitative results provide strong evidence for the effectiveness of our OPOCA.

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TABLE III
COMPARISON RESULTS ON SEMANTICPOSS TEST DATASET. OPOC MEANS ANNOTATING ONE POINT FOR ONE CLASS IN SINGLE LIDAR SCANS. † REPRESENT THE REPRODUCED RESULTS

Setting	Model	mIoU	people	rider	car	traffic sign	trunk	plants	pole	fence	building	bike	road
Full.Sup.	PointNet++ [22]	20.1	20.8	0.1	8.9	21.8	4.0	51.2	3.2	6.0	42.7	0.1	62.2
	SequeezeSegV2 [29]	29.8	18.4	11.2	34.9	11.0	15.8	56.3	4.5	25.5	47.0	32.4	71.3
	KPConv [49]	55.7	**79.2**	22.2	65.2	**30.8**	**75.1**	26.6	37.9	**83.4**	57.4	**54.4**	80.1
	RandLA-Net [8]	53.5	69.2	**26.7**	77.2	24.3	27.3	73.4	30.9	51.4	**82.5**	43.9	**81.2**
	MinkUNet14A† [26]	**56.8**	77.9	26.0	**82.3**	26.0	28.0	**74.0**	**43.4**	53.9	80.9	53.2	79.2
10%	KPConv [49]	53.5	76.5	23.4	62.9	29.4	**74.1**	17.5	**38.1**	82.1	54.6	49.8	80.1
	RandLA-Net [8]	49.4	67.1	24.6	**73.4**	18.6	23.5	**63.8**	29.9	51.7	**82.1**	40.6	**81.0**
	WS3 [50]	48.8	74.4	**30.9**	42.9	26.0	70.3	7.4	31.9	78.5	43.2	52.3	79.2
	PSD [51]	50.0	75.8	28.0	57.0	25.1	68.9	9.4	35.9	78.4	42.0	50.5	79.2
	Xu [52]	**54.0**	**78.6**	19.6	69.4	**30.6**	74.0	15.0	36.8	**82.3**	55.4	**52.5**	80.2
OPOC	KPConv [49]	41.1	59.8	12.3	45.6	7.2	23.0	60.7	22.6	31.1	74.6	42.2	72.9
	RandLA-Net [8]	36.4	59.0	10.0	18.5	1.9	20.1	50.7	22.9	29.4	68.6	44.5	**74.8**
	Baseline(MinkUNet14A†) [26]	46.0	**68.1**	23.3	60.6	7.0	20.9	62.3	31.3	33.8	75.7	51.4	71.7
	Ours(MinkUNet14A†)	**50.2**	63.8	**33.6**	**75.7**	**9.6**	**26.3**	**69.3**	**32.2**	**36.6**	**79.2**	**54.0**	71.5

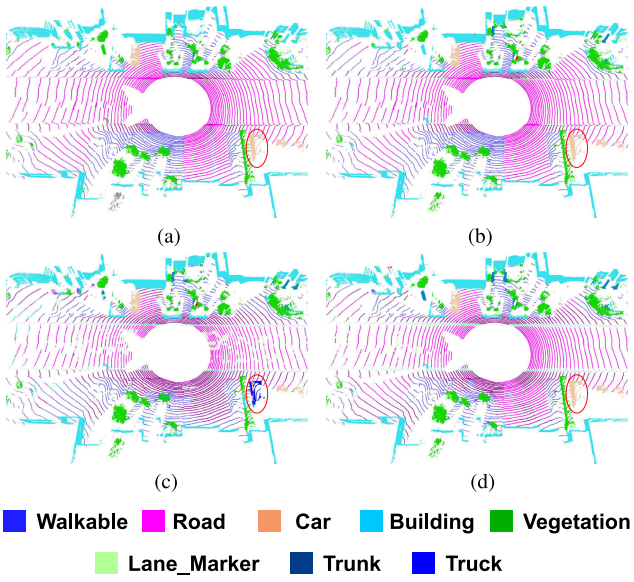


Fig. 6. Qualitative results achieved by (a) ground truth, (b) full-supervised, (c) baseline, and (d) our OPOCA on the WOD.

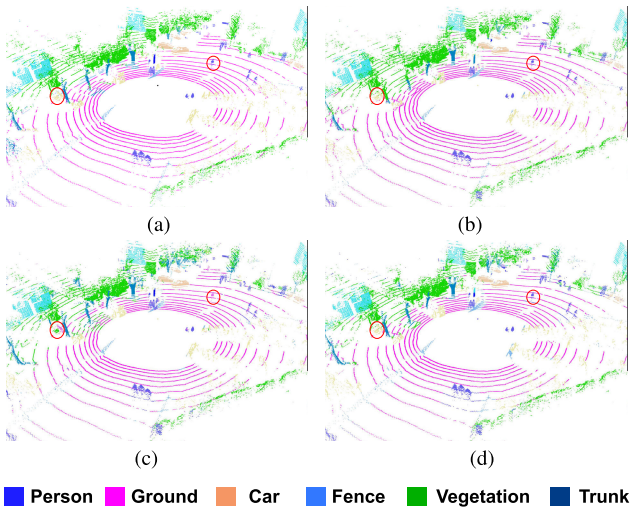


Fig. 7. Qualitative results achieved by (a) ground truth, (b) full-supervised, (c) baseline, and (d) our method on the SemanticPoss [16] dataset.

D. Results on SemanticPOSS

In this experiment, we compared our OPOCA with existing fully supervised methods, random percentage labels, and the MinkUNet14A backbone [26] on the SemanticPoss [16] validation set. In SemanticPOSS, the number of OPOCA accounts for approximately 0.192‰ of the total points. This is higher than the counts in the other two datasets, possibly due to the sparser nature of the point cloud in SemanticPOSS. As shown in Table III, our OPOCA improves the baseline by **4.2** mIoU. 9 out of the 11 classes show improvements compared to the baseline. These significant improvements across the majority of categories provide strong evidence for the effectiveness of our approach. In full supervision, both KPConv [49] and RandLA-Net [8] demonstrate comparable performance with our chosen backbone. However, under the OPOC labels, these two methods significantly performed worse than our approach, suggesting they are less suitable for situations involving extremely sparse annotations. The qualitative results (see Fig. 7) display the results achieved by ground truth, fully supervised methods, baseline, and our method. The areas where our approach excels are highlighted with red circles. This visual representation clearly demonstrates the effectiveness of our method.

E. Ablation Study

To validate the effectiveness of each component, we conducted an extensive ablation study on the SemanticKITTI dataset [9]. The baseline is the MinkUNet14A [26] model trained with the OPOCA strategy. We performed experiments with different combinations of the proposed components: PLSM, SDL, RIAM, and STM. The results of these experiments are shown in Table IV; a checkmark ✓ indicates that the component was adopted.

Effects of Operations in PLSM: According to Table V, after applying the PLSM (1/2/3) to three sets of randomly selected initial labeled points, the number of labeled points, mIoU, and accuracy are all very close. This indicates that our PLSM is not sensitive to the initial point selection and exhibits strong robustness. In addition, when removing ground filtering or normal vector constraints in PLSM 3, the quality of pseudo

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TABLE IV
ABLATION STUDY ON SEMANTICKITTI VALIDATION SET

Exp ID	PLSM	SDL	RIAM	STM	mIoU
1					43.0
2			✓		43.5
3	✓				41.4
4	✓	✓			47.2
5	✓	✓	✓		48.7
6	✓	✓		✓	49.9
7	✓	✓	✓	✓	50.4

TABLE V
ABLATION STUDY OF PLSM MODULE ON SEMANTICKITTI: RANDOM SELECTION OF INITIAL SEED POINTS (PLSM 1/2/3), NORMAL VECTOR CONSTRAINT (PLSM), AND GROUND FILTERING (RANSAC) (LIMITED TO POINTS WITH PSEUDO LABELS)

	Labeled Points	mIoU	acc
PLSM1	87714716	72.51	89.86
PLSM2	87776379	72.74	89.97
PLSM3	87777370	72.67	89.92
PLSM3 (w/o normal)	377949089	60.62	78.54
PLSM3 (w/o ransac)	103812741	72.48	87.89

labels decreases. This demonstrates that these operations in PLSM can improve the quality of pseudo labels and are necessary.

We conducted an ablation analysis of the PLSM module's hyperparameters on SemanticKITTI, as detailed in Table VI. The RANSAC parameters, including dis_{th} (maximum inlier distance from the plane model), ransac_n (initial inlier count per iteration), and iters (iteration count), were analyzed, along with the DBSCAN parameters eps (density parameter for neighboring point identification) and min_{pts} (minimum cluster size). Our findings reveal that adjustments to these parameters minimally impact the quality of pseudo labels derived from the PLSM module, highlighting the module's robustness and the flexibility of our hyperparameter settings.

Effects of PLSM and SDL: According to the information in Table IV, it is seen that using PLSM alone has a negative effect, resulting in a reduction of **1.6** mIoU compared to the baseline. We speculate that the accuracy decrease is due to the potential ambiguity and inaccuracy in the pseudo labels from naive spreading. At the same time, we believe that there are certain acceptable and unavoidable errors in pseudo-labeling based on geometric information. To mitigate the negative impact of erroneous pseudo labels, we propose SDL, which is designed to be used in conjunction with PLSM. As shown in Table IV, when PLSM and SDL are combined, there is an improvement of **4.2** mIoU compared to the baseline. This result effectively demonstrates the effectiveness of both components working together.

Effects of RIAM: In Table IV, it is seen that our proposed RIAM boosts the baseline by **0.5** mIoU and when the RIAM is added to the network structure based on the PLSM, mIoU is also improved by **1.5**. Furthermore, when combined with STM, RIAM increases the mIoU by **0.5**. This consistent improvement across different experiments with the addition of the module demonstrates its effectiveness and generalizability.

TABLE VI
ABLATION STUDY OF HYPERPARAMETERS IN THE PLSM MODULE ON SEMANTICKITTI: FOCUSING ON PARAMETERS FOR RANSAC AND DBSCAN (LIMITED TO POINTS WITH PSEUDO LABELS)

dis_{th}	ransac_n	iters	eps	min_{pts}	mIoU	acc
0.129	3	40	0.1	5	79.24	91.39
0.1					79.14	91.10
0.5					79.65	92.27
1					79.66	92.27
0.129	5				79.13	91.40
	7				78.97	91.37
	3	30			79.22	91.39
		50			79.25	91.42
		40	0.05		78.91	91.27
			0.2		79.73	91.46
			0.5		79.16	91.12
			0.1	3	79.64	91.56
				7	79.06	91.37

Effects of STM: As shown in Table IV, adding STM on top of PLSM and SDL is able to bring about a **2.7** mIoU improvement, whereas combining RIAM at the same time and adding STM on top of it will bring about a **1.7** mIoU improvement, which is a good proof of the effectiveness of STM.

V. CONCLUSION

We proposed a novel weakly supervised LiDAR PCSS method (OPOCA) with extremely few labels (OPOC), which only needs one point to be annotated for each class in single LiDAR scans. Specifically, two key components, PLSM uses prior geometric information and nonlearning methods to obtain a large number of pseudo-labels, which solves the scarcity of labels; RIAM integrates the features of other views into the backbone to improve the diversity of features. Extensive experiments on SemanticKITTI, WOD, and SemanticPoss datasets fully demonstrate our proposed module's effectiveness and our approach's rationality.

Limitations: OPOCA still needs to find the position of the points to be annotated for each class, and finding the position of the points of the category with fewer points will take up a large part of the labeling time. Further work may involve using only scene-level annotations to further reduce reliance on laborious annotation.

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